

Member ID: _____

Time: _____

Rank: _____



SQL DATABASE FUNDAMENTALS (345)

NATIONAL 2022

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each) _____ (100 points)

TOTAL POINTS _____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

MULTIPLE CHOICE

Identify the letter of the choice that *best* completes the statement or answers the question.

1. _____ is the name of a query that is placed within a WHERE clause.
 - a. Partial query
 - b. Master query
 - c. Sub query
 - d. Where query
2. In a relational database model, a tuple is commonly referred to as a _____.
 - a. table
 - b. column
 - c. row
 - d. relationship
3. Given `CREATE TEMPORARY TABLE t1 (c1 int primary key) ENGINE=INNODB`
What command would be used to find and display t1 - the name of the temp table?
 - a. `SHOW TABLES FROM INFORMATION_SCHEMA LIKE 'INNODB_TEMP%'`
 - b. `SELECT * FROM INFORMATION_SCHEMA.INNODB_TEMP_TABLE_INFO`
 - c. `SHOW TABLES WHERE TYPE(test) = TEMP`
 - d. The actual table name cannot be returned
4. Temporary tables created during a stored procedure persist until _____.
 - a. the stored procedure ends
 - b. a commit has completed on a transaction
 - c. it is manually deleted inside the stored procedure code
 - d. the user connection is successfully terminated
5. What is the purpose of port obfuscation?
 - a. To allow all RDBMs to utilize a standardized port
 - b. To alter the default port as an additional layer against targeted attacks
 - c. To ensure portability when utilizing multiple programming languages
 - d. To move a database from a local instance to the cloud
6. The standard endpoint Port for common database servers would be _____.
 - a. MySQL/Aurora - 3306, SQL Server - 1433, PostgreSQL - 5432, Oracle - 1521
 - b. MySQL/Aurora - 80, SQL Server - 443, PostgreSQL - 8000, Oracle - 5430
 - c. MySQL/Aurora - 2213, SQL Server - 6155, PostgreSQL - 4433, Oracle - 2204
 - d. MySQL/Aurora - 7100, SQL Server - 7100, PostgreSQL - 7100, Oracle – 7100

7. What is the difference between a super key and a candidate key?
 - a. A super key is a synonym for a primary key whereas one of the candidate keys can be a primary key
 - b. All super keys are candidate keys
 - c. All parts of a super key are necessary to uniquely identify a row, whereas not all parts of a candidate key are necessary to identify a unique row.
 - d. Super keys can contain null values and candidate keys cannot contain null values
8. An example of data redundancy would be _____.
 - a. Table of Users has a status_id and table of Administrators has a status_id
 - b. Table of Users has an entry for a John Jones and an entry in the Administrator table has the same name of John Jones
 - c. Table of Users has character state field and table of Administrators has a character state field
 - d. Table of Users contains the same exact columns as the Administrators table
9. Why would a database administrator choose to denormalize a database?
 - a. To avoid duplication of data
 - b. To save space
 - c. To make certain queries more efficient
 - d. They wouldn't, rules of normalization apply when designing databases
10. Which of the following best describes database transactional isolation?
 - a. The transaction only occurs in one process
 - b. The transactions occur independently without interference from other transactions
 - c. The entire transaction takes place after all others complete
 - d. The transaction must be consistent before and after the transaction
11. What is meant by database consistency?
 - a. Integrity constraints must be maintained so that the database is consistent before and after a transaction.
 - b. Atomicity constraints must be maintained so that the database is consistent before and after a transaction.
 - c. Isolation constraints must be maintained so that the database is consistent before and after a transaction.
 - d. None of these are correct
12. The four standard database Isolation levels are _____.
 - a. Read Uncommitted, Read Committed, Serializable, Repeatable Read
 - b. Dirty Read, Repeatable Dirty Read, Serializable, Phantom Read.
 - c. Dirty Read, Repeatable Dirty Read, Repeatable Read, Phantom Read.
 - d. Dirty Read, Uncommitted, Committed, Phantom Read.

13. Which of the following is not an example of a Transactional Phantom Read?
- When the same queries are executed, but the rows retrieved by the two are different
 - While in the course of a transaction, new rows are added or removed by another transaction to the records being read
 - When range locks are not acquired on performing a SELECT/WHERE operation while new rows are added or removed by another transaction
 - When two or more processes attempt to access the same data at the same time
14. Which of the following does *not* describe the Read Uncommitted Isolation Level?
- It is used to allow dirty reads
 - It is used to improve performance in high concurrency environments
 - It allows non-repeatable reads
 - It is used to lock a row so no other read can access the data
15. Optimistic concurrency control allows _____.
- Multiple transactions to run simultaneously.
 - Selecting data based on the system controller.
 - Commits or rollbacks based on conflict validation.
 - Performance improvements in high data contention environment
16. Concurrency should be implemented as _____.
- Optimistic or pessimistic
 - Optimistic and pessimistic
 - Semi-optimistic or semi-pessimistic
 - Semi-optimistic and semi-pessimistic
17. What is a pessimistic database lock?
- Lock is granted upon attempting to access the data; others can't access data until transaction is complete
 - Lock is granted upon update only; others access when update has completed
 - Used mostly when multiple transactions do not interfere with each other
 - Lock is granted after reading the data; others can't access data until transaction is complete
18. Commitment ordering is best described as a _____.
- specific set of interoperable serializability libraries in concurrency control of databases & transaction processing
 - language extension library of interoperable serializability methods in concurrency control of databases & transaction processing
 - class of interoperable serializability techniques in concurrency control of databases & transaction processing
 - way to ensure a specific isolation level is achieved upon transaction commits

19. The *most* important advantage of concurrency is _____.
a. isolation integrity
b. increased program throughput
c. high responsiveness for database mirroring
d. pessimistic isolation level
20. Which of the following is a *true* statement?
a. Native SQL stored procedures are written exclusively with SQL commands.
b. Native SQL stored procedures can be written in a language such as Java, C, and JavaScript.
c. Stored procedures are only supported in major databases.
d. Stored procedures are compatible across databases.
21. _____ is an advantage of a stored procedure.
a. Maintainability
b. Localizes business rules
c. Ease of development
d. Portability
22. What would be the definition of a natural join?
a. A join between two tables where the data returned is restricted to key data.
b. A join between two tables where columns are defined and named the same.
c. A join where data types of the keys between the two tables have differing data types.
d. A join where the data returned does not contain any orphaned data.
23. What is the key difference between SQL and NoSQL?
a. SQL is associated with relational databases and NoSQL are not relational
b. NoSQL uses standard programming languages to access the database instead of the SQL standard
c. NoSQL is modeled using tabular relations
d. SQL is used in *big data* applications and NoSQL does not meet the needs of *big data*
24. _____ is a DML command.
a. COMMIT
b. ROLLBACK
c. SAVEPOINT
d. SELECT
25. Which of the following is *true* about an E-R Model?
a. Current tools to display are not complete
b. It defines the conceptual view of the database
c. Best to create for business after database is in production
d. It's an outdated method of designing databases

26. In an E-R diagram, the degree of a relationship is most often represented by which relationships?
- a. Connectivity, cardinality
 - b. One-one, one-many, many-one
 - c. Subtype, supertype
 - d. Binary, ternary, unary
27. An example of a schema in a database would be a _____.
a. relational
b. tree
c. logical
d. modular
28. In an E-R diagram, the connectivity of a relationship is *most* often represented by which relationships?
- a. Binary, ternary, unary
 - b. One-one, one-many, many-one
 - c. Subtype, supertype
 - d. Connectivity, cardinality
29. An example of a binary relationship would be _____.
a. a teacher teaches a subject
b. a subject is a prerequisite for another subject
c. a teacher teaches a course which defines a subject
d. the same subject is taught in multiple courses
30. What is an implicit cursor (Oracle)?
a. An implicit cursor changes the consistent state of the database to another state
b. A temporary table
c. Implicit cursors are declared automatically when running complex queries that include subqueries
d. A data structure used between two connected databases
31. A difference between a clustered and a non-clustered index is a _____.
a. clustered index involves multiple tables and a non-clustered is one table
b. clustered index re-orders the way the records in the tables are stored and a non-clustered does not
c. clustered index optimizes the data with multiple keys and a non-clustered uses a single key
d. clustered index stores data in one location and indices in another and a non-clustered index stores them together

32. What is *not* a property of a data dictionary in database terms?

- a. Holds the names of all the tables and their schemas
- b. It lists security constraints on the tables
- c. Describes where and how tables are stored
- d. It's a graphical representation of the data model

33. The following two queries will produce the same result.

SELECT * FROM table WHERE item IN (9,10,11,12)

SELECT * FROM table WHERE item BETWEEN (8, 13)

- a. True
- b. False

Consider the following three tables for questions 34-41

Member's MemberId is auto-increment, unique; Item's Id is auto-increment, unique; All fields are non-null

Member				Member_Item		Item			
memberId	fName	lName	status	memberId	itemId	id	descrip	paid	size_cm
1004	Jim	Jones	Active	1117	50	30	Item1	33.50	1102
1117	Jan	Smith	Retired	1551	30	40	Item2	66.25	5
1551	Betty	Shine	Active	1551	55	50	Item3	80.80	5512
				1004	30	55	Item3a	17.71	3232
				1551	75	70	Item5	52.31	16
				1117	30	75	Item5a	100.65	505

34. What would not be the appropriate constraints for the Member_Item table?

- a. FOREIGN KEY (memberId) REFERENCES Member(memberId)
- b. FOREIGN KEY (itemId) REFERENCES Item(id)
- c. PRIMARY KEY (memberId) FOREIGN KEY(itemId)
- d. PRIMARY KEY (itemId) FOREIGN KEY (memberId)

35. How many rows will be returned from the following query?

```
SELECT memberId
  FROM Member a
        INNER JOIN Member_Item b on a.memberId = b.memberId
        INNER JOIN Item c on b.itemId = c.id
 WHERE paid < 52.31
```

- a. 5
- b. 3
- c. 1
- d. 0

36. What level of normalization is the Member table?

- a. 1NF, 2NF, 4NF, 5NF (not 3NF)
- b. 1NF, 2NF, 3NF, 4NF
- c. 1NF, 2NF, 3NF
- d. 1NF, 2NF

37. How many rows will be returned from the following query?

```
SELECT a.memberId
  FROM Member a
        INNER JOIN Member_Item b on a.memberId = b.memberId
        INNER JOIN Item c on b.itemId = c.id
 WHERE paid < (SELECT AVG(paid) FROM Item)
```

- a. 5
- b. 4
- c. 3
- d. 2

38. What will the result be from the following query?

```
SELECT MAX(size_cm), MIN(size_cm)
  FROM Member a
        INNER JOIN Member_Item b on a.memberId = b.memberId
        RIGHT JOIN Item c on b.itemId = c.id
 WHERE paid < (SELECT SUM(paid) / COUNT(size_cm) FROM Item)
```

- a. 5512, 5
- b. 5512, 16
- c. 3232, 5
- d. 3232, 16

39. What statement would change the datatype of the size_cm field from int to smallint?
- ALTER TABLE Item CHANGE COLUMN size_cm size_cm smallint
 - UPDATE TABLE Item SET DATA_TYPE = "smallint" WHERE COLUMN_NAME = size_cm
 - ALTER TABLE Item CHANGE size_cm FROM int TO smallint
 - ALTER TABLE Item MODIFY size_cm DATA_TYPE = smallint
40. What would the joins look like to show them items not associated with any members?
- SELECT fName, lname, descrip
- FROM Member a
- < Fill in the JOINS here>
- WHERE fname is NULL
- INNER JOIN Member_Item b on a.memberId = b.memberId
INNER JOIN Item c on b.itemId = c.id
 - INNER JOIN Member_Item b on a.memberId = b.memberId
RIGHT JOIN Item c on b.itemId = c.id
 - LEFT JOIN Member_Item b on a.memberId = b.memberId
RIGHT JOIN Item c on b.itemId = c.id
 - RIGHT JOIN Member_Item b on a.memberId = b.memberId
RIGHT JOIN Item c on b.itemId = c.id
41. What would the following statement produce?
- INSERT INTO Item VALUES (75, "Item 5a", 202.20, 1212);
- Id 75 will be replaced with the record being added
 - A duplicate item will be added with an id of 75
 - A new item will be inserted with an id of 76
 - A message about record being overwritten (If warning settings are set to warn)
42. Under what situation will an index not have a positive impact on performance?
- When a database is has a small amount of tables
 - During a data import
 - During a query that accesses the index
 - When table join includes multiple tables
43. Which of the following is not a measure a developer can use to avoid SQL Injection?
- Build parameterized queries
 - Use stored procedures
 - By having one entry point to the database
 - Restrict administrative access to database

44. What is a major difference between MySQL and Microsoft's SQL Server?
- SQL Server is designed for high scalability; MySQL's major market is companies with less than a million records
 - SQL Server is proprietary; MySQL is open source
 - SQL Server supports more platforms than MySQL
 - MySQL does not use E-R model tools and SQL Server does

Consider the following for questions 45-47 (MySQL).

DROP TABLE IF EXISTS testTable;

```
CREATE TABLE `testTable` (  
  `id` INT NOT NULL AUTO_INCREMENT,  
  `fName` VARCHAR(45) NOT NULL,  
  `lName` VARCHAR(45),  
  `dob` DATETIME NOT NULL,  
  PRIMARY KEY (`id`));
```

45. How would you add an index on last name followed by first name to the table above?
- USE testTable ADD INDEX fullName (lName, fName)
 - ALTER TABLE testTable ADD INDEX fullName (lName, fName)
 - MODIFY testTable ADD SECONDARY fullName (lName, fName)
 - INDEX ON (lName, fName) INTO testTable
46. What would the query be to show the table information (name, column, datatypes, etc.) for testTable?
- SELECT Columns FROM TestDB WHERE TABLE = "testTable"
 - SELECT Schema.* FROM TABLE_INFORMATION WHERE TABLE_NAME = "testTable"
 - SELECT * FROM INFORMATION_SCHEMA.COLUMNS WHERE TABLE_NAME = "testTable"
 - SELECT * FROM SCHEMA WHERE TABLE_NAME = "testTable"
47. To import a CSV file into this table from the command line you would use _____.
- LOAD DATA LOCAL INFILE 'file.csv' INTO TABLE testTable FIELDS TERMINATED BY ',' ENCLOSED BY '"' LINES TERMINATED BY '\n'
 - IMPORT 'file.csv' INTO testTable
 - ADD DATA TO testTable FROM testTable WITH (',', '\n')
 - UPDATE testTable FROM 'file.csv' DEFAULT

48. Which of the following would not be considered a typical database index type?
- a. Bitmap
 - b. B-tree
 - c. Alpha
 - d. Clustered
49. Given tables *CarCustomers* and *PaintCustomers*, which query would not get you all the unique first names that exist in both tables?
- a.

```
SELECT DISTINCT first_name FROM
(SELECT first_name FROM CarCustomers
UNION
SELECT first_name FROM PaintCustomers) names
```
 - b.

```
SELECT DISTINCT a.first_name FROM CarCustomers a
INNER JOIN PaintCustomers b ON a.first_name = b.first_name
WHERE b.first_name IS NULL
UNION
SELECT DISTINCT b.first_name FROM CarCustomers a
RIGHT JOIN PaintCustomers b ON a.first_name = b.first_name
WHERE a.first_name IS NULL
UNION
SELECT DISTINCT a.first_name FROM CarCustomers a
INNER JOIN PaintCustomers b ON a.first_name = b.first_name
```
 - c.

```
SELECT DISTINCT a.first_name, b.first_name FROM CarCustomers a,
PaintCustomers b
WHERE a.first_name = b.first_name
```
 - d.

```
SELECT first_name FROM CarCustomers
UNION
SELECT first_name FROM PaintCustomer
```
50. Given two databases *CarSales* and *PaintSales* existing on the same database server, what can be said about the following query?
- ```
SELECT title, sales FROM
(SELECT "Car" as title, SUM(sales) as sales FROM CarSales.sales
UNION
SELECT "Paint" as title, SUM(paintsSold) as sales FROM PaintSales.sales)
```
- a. It is not possible to query between databases
  - b. When using UNIONS the columns that are selected must be the same - this query will not run
  - c. The query will produce an error stating the derived table needs an alias
  - d. The query will run as expected